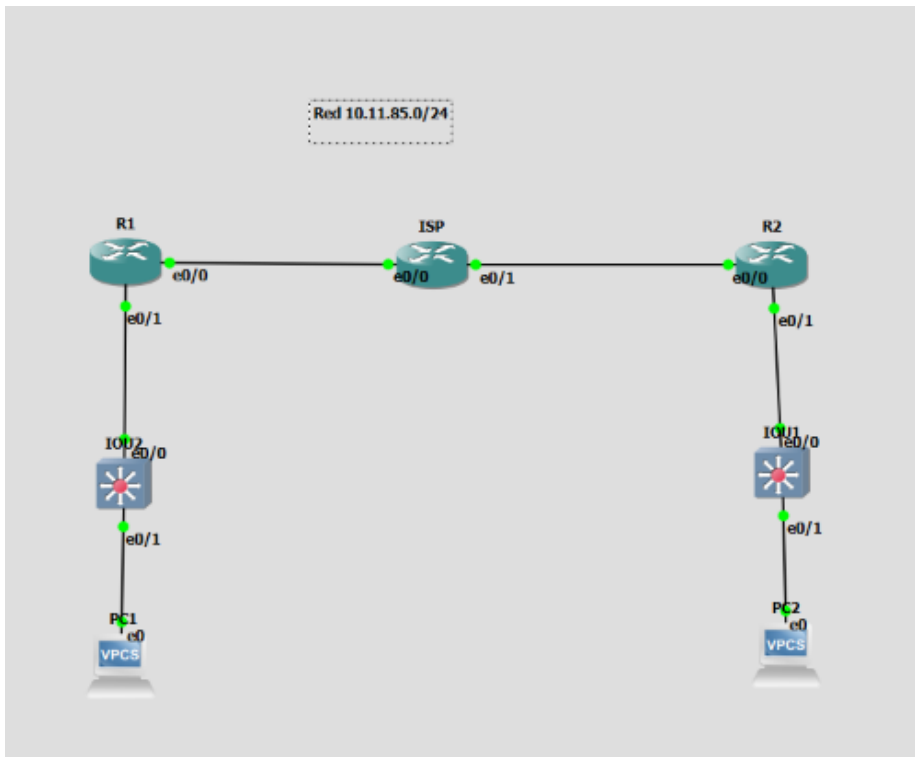


Objetivo

Implementar **GRE over IPSec con IKEv2**. Combina GRE (soporte multicast, enrutamiento dinámico) con IKEv2 (negociación en 4 mensajes, DPD nativo). IPSec en modo **transport** cifra el payload GRE. La diferencia vs IKEv1 GRE es únicamente la sección IKE: se añade **set ikev2-profile** en el crypto map.

Topología



Encapsulación

Capa	Contenido
IP original	src 10.11.85.2 dst 10.11.85.194 (PC1→PC2)
GRE	Tunnel0 encapsula con IP router: .65 → .130
IPSec ESP (transport)	Cifra payload GRE con AES-256 / SHA256
IP outer WAN	src 10.11.85.65 dst 10.11.85.130

Parametros

Parámetro	Valor
Transform Set	TS-GRE-IKEV2
Modo IPSec	TRANSPORT (GRE ya encapsula)
Cifrado / Integridad	AES-256 / HMAC-SHA-256
Parámetro	Valor
Proposal	PROP-IKEV2
Cifrado	AES-CBC-256
Integridad	SHA-256
Grupo DH	14 (2048-bit)
Keyring / PSK	KR-IKEV2 / ITLA2024
Profile	PROF-IKEV2
Auth local/remota	Pre-Shared Key
ACL tráfico	permit gre host 10.11.85.65 host 10.11.85.130

Crypto Map	CM-GRE-IKEV2 + set ikev2-profile PROF-IKEV2
Tunnel mode	gre ip

Configuración R1

```

enable
configure terminal
hostname R1
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.65 255.255.255.192
no shutdown

interface Ethernet0/1
description LAN hacia SW1-PC1
ip address 10.11.85.1 255.255.255.192
no shutdown
ip route 0.0.0.0 0.0.0.0 10.11.85.66

interface Tunnel0
description GRE Tunnel hacia R2 (IKEv2)
ip address 10.10.10.1 255.255.255.252
tunnel source Ethernet0/0
tunnel destination 10.11.85.130
tunnel mode gre ip
no shutdown

ip route 10.11.85.192 255.255.255.192 Tunnel0
crypto ikev2 proposal PROP-IKEV2
encryption aes-cbc-256
integrity sha256
group 14
crypto ikev2 policy POL-IKEV2
proposal PROP-IKEV2
crypto ikev2 keyring KR-IKEV2
peer R2
address 10.11.85.130
pre-shared-key ITLA2026
!
crypto ikev2 profile PROF-IKEV2
match identity remote address 10.11.85.130 255.255.255.255
authentication remote pre-share
authentication local pre-share

```

```
keyring local KR-IKEV2
crypto ipsec transform-set TS-GRE-IKEV2 esp-aes 256 esp-sha256-hmac
mode transport
ip access-list extended ACL-GRE-VPN
permit gre host 10.11.85.65 host 10.11.85.130
crypto map CM-GRE-IKEV2 10 ipsec-isakmp
set peer 10.11.85.130
set transform-set TS-GRE-IKEV2
set ikev2-profile PROF-IKEV2
match address ACL-GRE-VPN
interface Ethernet0/0
crypto map CM-GRE-IKEV2

end
```

Configuración R2

```
enable
configure terminal

hostname R2

interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.130 255.255.255.192
no shutdown

interface Ethernet0/1
description LAN hacia SW2-PC2
ip address 10.11.85.193 255.255.255.192
no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.129

interface Tunnel0
description GRE Tunnel hacia R1 (IKEv2)
ip address 10.10.10.2 255.255.255.252
tunnel source Ethernet0/0
tunnel destination 10.11.85.65
tunnel mode gre ip
no shutdown

ip route 10.11.85.0 255.255.255.192 Tunnel0
```

*crypto ikev2 proposal PROP-IKEV2
encryption aes-cbc-256
integrity sha256
group 14*

*crypto ikev2 policy POL-IKEV2
proposal PROP-IKEV2*

*crypto ikev2 keyring KR-IKEV2
peer R1
address 10.11.85.65
pre-shared-key ITLA2026
!*

*crypto ikev2 profile PROF-IKEV2
match identity remote address 10.11.85.65 255.255.255.255
authentication remote pre-share
authentication local pre-share
keyring local KR-IKEV2*

*crypto ipsec transform-set TS-GRE-IKEV2 esp-aes 256 esp-sha256-hmac
mode transport*

*ip access-list extended ACL-GRE-VPN
permit gre host 10.11.85.130 host 10.11.85.65*

*crypto map CM-GRE-IKEV2 10 ipsec-isakmp
set peer 10.11.85.65
set transform-set TS-GRE-IKEV2
set ikev2-profile PROF-IKEV2
match address ACL-GRE-VPN*

*interface Ethernet0/0
crypto map CM-GRE-IKEV2*

end

Configuración ISP

*enable
configure terminal
hostname ISP*

```

interface Ethernet0/0
ip address 10.11.85.66 255.255.255.192
no shutdown

interface Ethernet0/1
ip address 10.11.85.129 255.255.255.192
no shutdown

end
write memory

```

Verificación

! Estado SA IKEv2 (QM_IDLE → READY en IKEv2)

```
show crypto ikev2 sa
```

```

R1#show crypto ikev2 sa
IPv4 Crypto IKEv2  SA

Tunnel-id Local          Remote          fvr/fivr/f   Status
1          10.11.85.65/500      10.11.85.130/500  none/none    READY
    Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/14 sec

IPv6 Crypto IKEv2  SA

```

```
show crypto ikev2 sa detailed
```

```

R2#show crypto ikev2 sa detailed
IPv4 Crypto IKEv2  SA

Tunnel-id Local          Remote          fvr/fivr/f   Status
1          10.11.85.130/500      10.11.85.65/500  none/none    READY
    Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
    Life/Active Time: 86400/83 sec
    CE id: 1001, Session-id: 1
    Status Description: Negotiation done
    Local spi: 97FFAA0ABCC97057      Remote spi: 598678D16559D325
    Local id: 10.11.85.130
    Remote id: 10.11.85.65
    Local req msg id: 0               Remote req msg id: 2
    Local next msg id: 0              Remote next msg id: 2
    Local req queued: 0               Remote req queued: 2
    Local window: 5                   Remote window: 5
    DPD configured for 0 seconds, retry 0
    Fragmentation not configured.
    Extended Authentication not configured.
    NAT-T is not detected
    Cisco Trust Security SGT is disabled
    Initiator of SA : No

IPv6 Crypto IKEv2  SA

```

```
show crypto ikev2 proposal
```

```

R1#show crypto ikev2 proposal
IKEv2 proposal: PROP-IKEV2
  Encryption : AES-CBC-256
  Integrity   : SHA256
  PRF         : SHA256
  DH Group    : DH_GROUP_2048_MODP/Group 14
IKEv2 proposal: default
  Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
  Integrity   : SHA512 SHA384 SHA256 SHA96 MD596
  PRF         : SHA512 SHA384 SHA256 SHA1 MD5
  DH Group    : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
R1#

```

! SA IPsec + contadores encrypt/decrypt y selectores proto 47 (GRE) confirman que IPsec protege el GRE

show crypto ipsec sa

```

R1#sh crypto ipsec sa
interface: Ethernet0/0
  Crypto map tag: CM-GRE-IKEV2, local addr 10.11.85.65

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.11.85.65/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (10.11.85.130/255.255.255.255/47/0)
  current peer 10.11.85.130 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.11.85.65, remote crypto endpt.: 10.11.85.130
    plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/0
    current outbound spi: 0x0(0)
    PFS (Y/N): N, DH group: none

  inbound esp sas:
    spi: 0xAB94F162(2880762210)
      transform: esp-256-aes esp-sha256-hmac ,
      in use settings ={Transport,}
      conn id: 1, flow_id: SW:1, sibling_flags 80000000, crypto map: CM-GRE-IKEV2
      sa timing: remaining key lifetime (k/sec): (4228447/3549)
      IV size: 16 bytes
      replay detection support: Y
      ecn bit support: Y status: off
      Status: ACTIVE(ACTIVE)

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
    spi: 0xB56578D1(3043326161)
      transform: esp-256-aes esp-sha256-hmac ,
      in use settings ={Transport,}
      conn id: 2, flow_id: SW:2, sibling_flags 80000000, crypto map: CM-GRE-IKEV2
      sa timing: remaining key lifetime (k/sec): (4228447/3549)
      IV size: 16 bytes
      replay detection support: Y
    #pkts encaps: 9, #pkts encrypt: 9, #pkts digest: 9
    #pkts decaps: 9, #pkts decrypt: 9, #pkts verify: 9
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.11.85.130, remote crypto endpt.: 10.11.85.65
    plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/0
    current outbound spi: 0xB56578D1(3043326161)
    PFS (Y/N): N, DH group: none

```

```
show crypto ikev2 policy
```

```
R2#show crypto ikev2 policy
```

```
IKEv2 policy : POL-IKEV2  
  Match fvrfl : global  
  Match address local : any  
  Proposal    : PROP-IKEV2
```

```
IKEv2 policy : default  
  Match fvrfl : any  
  Match address local : any  
  Proposal    : default
```

```
show ip access-lists ACL-GRE-VPN con
```

```
el crypto session
```

```
AAA user authorization: none
```

```
R2#show crypto session
```

```
Crypto session current status
```

```
Interface: Ethernet0/0
```

```
Profile: PROF-IKEV2
```

```
Session status: UP-ACTIVE
```

```
Peer: 10.11.85.65 port 500
```

```
Session ID: 1
```

```
IKEv2 SA: local 10.11.85.130/500 remote 10.11.85.65/500 Active
```

```
IPSEC FLOW: permit 47 host 10.11.85.130 host 10.11.85.65
```

```
Active SAs: 2, origin: crypto map
```

```
Ping y trace, verificación del salto
```

```
PC2> ping 10.11.85.2
```

```
84 bytes from 10.11.85.2 icmp_seq=1 ttl=62 time=4.871 ms  
84 bytes from 10.11.85.2 icmp_seq=2 ttl=62 time=2.039 ms  
84 bytes from 10.11.85.2 icmp_seq=3 ttl=62 time=1.955 ms  
84 bytes from 10.11.85.2 icmp_seq=4 ttl=62 time=1.586 ms  
84 bytes from 10.11.85.2 icmp_seq=5 ttl=62 time=1.457 ms
```

```
PC2> trace 10.11.85.2
```

```
Trace to 10.11.85.2, 8 hops max, press Ctrl+C to stop
```

```
 1  10.11.85.193  0.918 ms  0.419 ms  1.823 ms  
 2  10.10.10.1   3.457 ms  0.667 ms  1.287 ms  
 3  *10.11.85.2  1.261 ms (ICMP type:3, code:3, Destination port unreachable)
```